GOVERNMENT COLLEGE OF ENGINEERING, CHHATRAPATI SAMBHAJINAGAR

CouncilConnect — Frontend Report

AI-Augmented Campus Governance & Operations Ecosystem · Final Year Project · Frontend (React.js) Implementation

PROGRESS REPORT · PROTOTYPE COMPLETE

1. What this report covers

This document summarises everything built so far for the **CouncilConnect frontend** — a single responsive web application for both PC and mobile. It explains the product flow in the order a user would experience it, lists every screen and feature, shows how to run the demo, and documents the handover point for the backend team.

Current status: All 15 screens are built, wired, and verified rendering without errors on every route. Two functional prototype gaps were found and fixed during this cycle (missing component imports and a broken toast hook). All interactive actions (event creation, lost & found posting, profile editing, position handover) now work against an in-browser data store.

2. Tech stack

LAYER	CHOICE
Framework	React 18 + Vite (single-page app, no native/monorepo tooling)
Routing	React Router (15 client-side routes)
Maps	Leaflet + OpenStreetMap (campus navigation)
Icons	Font Awesome (free CDN)
State / data	localStorage-persisted in-memory stores (`store.js` , `usersStore.js`) behind a mock API layer
Theme	Matched to the GECA T&P website: blue `#2266ED` , gold `#c8963e` , Fraunces + Inter fonts
Build output	`dist/` — JS bundle ~418 KB (gzip ~123 KB), CSS ~24 KB

3. How to run the demo

1. **Start the dev server**
   `cd frontend` → `npm run dev` (Node at `/usr/local/bin`)
2. **Open the app** — `http://localhost:5173/` (fixed port via `strictPort`)
3. **Log in** with any demo account below, or register as a student
4. **Production build** — `npm run build`

Demo accounts (shown on the Login page)

ROLE	EMAIL	PASSWORD	REDIRECTS TO
Student	`student@geca.ac.in`	`student`	Home
General Secretary (rep)	`cse_gs@geca.ac.in`	`gs`	Inbox
Branch Secretary (rep)	`ee_bs@geca.ac.in`	`rep`	Inbox

Admin / HOD	`cse_hod@geca.ac.in`	`admin`	Admin Dashboard
Hostel Rector	`rector@geca.ac.in`	`rector`	Admin Dashboard
Principal	`principal@geca.ac.in`	`principal`	Admin Dashboard

4. Product flow (how it works)

ENTRY & IDENTITY

A visitor lands on the **Home** page, which introduces CouncilConnect, shows live campus stats and quick links to every module. Students **Register** with a roll number (their address becomes `rollno@geca.ac.in`) or **Log in** with an institutional email. Representatives can optionally register *for a position*; the page live-previews their permanent role inbox, e.g. picking IT + General Secretary shows `it_gs@geca.ac.in`.

GOVERNANCE & COMMUNICATION

The **Council Directory** lists verified student representatives (search + branch + position filters, verification badges, term). Clicking a member opens their **Rep Profile**, which shows their verified role and a contact workflow (request → AI categorisation → approval → structured DM). Approved conversations live in **Messages**, which also shows incoming contact requests with approve / reject actions for representatives and admins.

EVENTS

Students **browse and register** for events and save favourites. Verified representatives and admins get a **Create Event** button; the form captures title, category, date, time, venue, capacity, poster and description. Publishing appends the event to the list immediately. A student who opens the create-event URL directly sees an *Access restricted* screen.

GRIEVANCES

Students **file complaints** (title, description, room, optional image). The form simulates **AI categorisation** and priority assignment, and the detail page shows a live status timeline (submitted → categorised → assigned → resolved / escalated). Representatives and admins see the **queue view** instead of their own complaints.

LOST & FOUND

Anyone can **post a lost or found item** (type, title, description, location, optional image). Found posts automatically get an *AI Match* badge; claims are sent to the poster. The list is filterable by All / Lost / Found and newly posted items appear instantly.

CAMPUS MAP

An **interactive Leaflet/OSM map** pins departments, hostels, event venues, offices and lost-item locations, with a type filter.

ADMIN & ROLE MANAGEMENT

Admins (HOD, Hostel Rector, Principal) land on the **Governance Dashboard** with analytics, a verification chain, escalations, an audit log, and a new **Positions** tab. The Positions tab lists every post with its **permanent role inbox** and current holder, and lets an admin **transfer** a post to a registered student — the role email stays, the previous holder reverts to a student account, and the new holder inherits the position.

PROFILE

Any user can **edit their own profile** (name, and for students branch / year / roll). For a post holder, the role inbox is shown as permanent and cannot be changed; the name shown is simply whoever currently holds the post.

5. Every screen at a glance

ROUTE	SCREEN	WHO CAN USE IT	KEY ACTIONS
`/`	Home	Everyone	Hero, stats, quick links to all modules
`/login`	Login	Everyone	Sign in, one-click demo accounts
`/register`	Register	Everyone	Student or rep registration + role-email preview
`/council`	Council Directory	Everyone	Search / filter representatives
`/council/:id`	Rep Profile	Everyone	View verified role, send contact request
`/messages`	Messages	Students, reps, admins	Threads, contact requests (approve/reject)

`/events`	Events	Everyone	Browse, register, save events
`/events/new`	Create Event	Reps & admins	Publish a new event
`/grievances`	Grievances	Students (own) / staff (queue)	File & track complaints
`/grievances/new`	New Grievance	Students	Submit with AI category preview
`/grievances/:id`	Grievance Detail	Everyone	Status timeline, escalation
`/lost-found`	Lost & Found	Everyone	Post / claim items
`/campus-map`	Campus Map	Everyone	Leaflet map with pins + filters
`/admin`	Admin Dashboard	Admins (HOD / Rector / Principal)	Analytics, verify, escalations, handover, audit
`/profile`	My Profile	Signed-in users	Edit name/details, view role inbox

6. Role-based emails & the “role stays, person changes” model

Every post has a **permanent forwarding address** in the form `{dept}_{post}@geca.ac.in` (e.g. `cse_gs@geca.ac.in`, `it_bs@geca.ac.in`). Institution-wide posts (Principal, Hostel Rector) use a simpler address: `principal@geca.ac.in`, `rector@geca.ac.in`.

What this means:

- The address is *not* tied to a person — it belongs to the post.
- When the holder changes, the address stays the same; mail is forwarded to whoever holds the post.
- The **Position Handover** screen makes this explicit: pick a post, pick a new holder, and the system moves the position (old holder → student, new holder → inherits the role email).
- An **Edit Profile** flow lets the current holder update their name at any time — no need to create new addresses when people leave or are re-elected.

Demo mapping of role emails

POST	ROLE INBOX	CURRENT HOLDER (DEMO)	BRANCH
General Secretary	`cse_gs@geca.ac.in`	Rohit Sharma	CSE
Branch Secretary	`ee_bs@geca.ac.in`	Priya Nair	Electrical
HOD / Faculty Admin	`cse_hod@geca.ac.in`	Dr. A. V. Kulkarni	CSE
Hostel Rector	`rector@geca.ac.in`	Dr. R. S. Patil	—
Principal	`principal@geca.ac.in`	Dr. V. R. Kulkarni	—

Note: the branch prefix in a role email reflects where the post currently sits; the handover keeps the existing address, matching the “address never changes” rule.

7. Code structure (where things live)

PATH	PURPOSE
`src/App.jsx`	Router with all 15 routes + AuthProvider / ToastProvider / ErrorBoundary / AppShell
`src/pages/`	One file per screen (listed in section 5)
`src/components/ui.jsx`	Reusable primitives: Button, Card, Badge, Chip, Table, Modal, Field, Tabs, EmptyState, Timeline, Reveal, StatBand, SearchInput, ToastProvider/useToast
`src/components/layout/`	Header, Navbar (hamburger on mobile), Footer, AppShell with scroll-progress bar
`src/context/AuthContext.jsx`	Login / register / logout; live user always read from the users store
`src/api/client.js`	**Mock API layer** — the single swap point for the real backend
`src/data/store.js`	In-memory store for events and lost & found items (with subscribe)
`src/data/usersStore.js`	Persisted users store: authentication, profile edits, position handover
`src/data/mock.js`	Mock datasets: council members, events, grievances, lost & found, map pins, analytics + `roleEmail()`
`src/theme/` + `src/styles/`	Design tokens and the three stylesheets (global / layout / components)

8. What's next

- **Backend handover:** replace the mock API in `src/api/client.js` and the two data stores with real endpoints — the screens do not need to change.

- **Position hierarchy rules:** wire event-creation permissions per your hierarchy (dept-level events → GS, college-wide → President) once the full hierarchy is finalised.
- **Polish:** animated counters, hero carousel, i18n (EN / Marathi / Hindi), attendance tracking.